

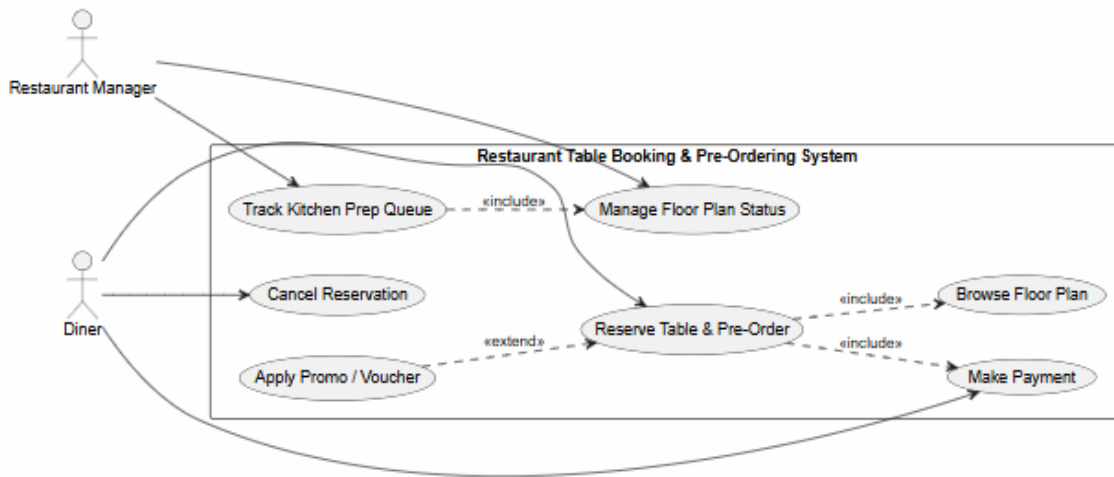
PROBLEM STATEMENT #36 | Retail, E-Commerce & Finance Restaurant Table Booking & Pre-Ordering App

Deliverable-1: Complete Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Table Selection & Booking	The system shall allow diners to select an available table from an interactive 2D floor plan for a specified date and time slot.	High	Pass: Selected table is locked and marked reserved for the duration. Fail: Double-booking or reservation overlap during the same time window.	Ensures diners secure their preferred dining spot without reservation conflicts.
FR-002	Menu Pre-Ordering	The system shall allow diners to browse the digital menu, customize items, and link pre-ordered dishes to their table reservation.	High	Pass: Item list and special notes are successfully attached to the reservation ID. Fail: Order items missing or unlinked from the booking.	Reduces dining wait times upon arrival and assists kitchen stock planning.
FR-003	Timeline & ETA Calculation	The system shall automatically compute estimated food prep times based on order volume and schedule kitchen preparation to align with the diner's expected arrival time.	Medium	Pass: Kitchen prep ticket triggers at (Arrival Time - Total Prep Time). Fail: Food is prepared too early (cold) or delayed beyond arrival	Ensures fresh, hot food is served immediately as diners are seated.
FR-004	Floor Management & Status Update	The system shall enable the Restaurant Manager to view live floor occupancy and manually or automatically update table statuses (<i>Available, Reserved, Seated, Billed, Cleaning</i>).	High	Pass: Floor plan reflects real-time status changes instantly across manager terminals. Fail: Terminal displays stale table status	Gives staff operational control over seating turnover and walk-ins.
FR-005	Advance Payment & Deposit	The system shall process advance deposits or full pre-order bill payments through an integrated payment gateway prior to booking confirmation.	Medium	Pass: Transaction completes, invoice generates, and reservation transitions to <i>Confirmed</i> . Fail: Payment failure leaves table unreserved.	Minimizes no-show rates and secures revenue for pre-prepared food.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Latency	Floor plan table status transitions must synchronize across all active manager and diner interfaces in under 500 ms.	High	Pass: WebSocket/push latency benchmarks $\leq$ 500 ms under 200 concurrent active users. Fail: Sync latency exceeds 500 ms.	Prevents concurrent conflicting bookings on the same table.
NFR-002	Security & Compliance	All customer payment details and personally identifiable information (PII) must be encrypted using AES-256 at rest and TLS 1.3 in transit, complying with PCI-DSS standards.	High	Pass: Automated security scans confirm zero unencrypted transmission of card/PII data. Fail: Plaintext storage or transmission detected.	Protects sensitive financial and personal data against interception and breach.

Deliverable-2:UML Diagram



Deliverable-3:Use Case Flow

Use Case ID: UC-01

Use Case Name: Reserve Table and Pre-Order Food

Primary Actor: Diner

Secondary Actor: Restaurant Manager / Kitchen Display System

Preconditions

1. Diner logged in with verified account.
2. Floor plan loaded with real-time table availability.
3. Kitchen menu synchronized (items, pricing, prep-time metadata).
4. Diner has valid payment method ready.
5. Selected date/time is within the restaurant's booking window.

Postconditions (Success)

1. Table status updated to "Reserved."
2. Pre-order kitchen ticket queued with itemized dishes and prep timeline.
3. Diner receives confirmation with QR code, table number, and ready-by time.
4. Payment authorized per restaurant policy.

Main Success Scenario (MSS):

1. Diner selects the target restaurant, dining date, arrival time, and party size.
2. System fetches and displays the interactive 2D floor layout showing available tables in real time (<<include>> Browse Floor Plan).
3. Diner taps an available table corresponding to their party size.
4. System temporarily locks the table for 10 minutes and presents the digital food menu.
5. Diner selects food items, specifies dietary customization/notes, and confirms the pre-order cart.
6. System calculates the total preparation timeline and subtotal.
7. Diner proceeds to checkout and enters payment details (<<include>> Make Payment).
8. System validates payment, marks the table as *Reserved*, generates a booking ID, and notifies the kitchen system.
9. System displays the reservation confirmation screen with booking summary and QR pass.

Alternate Flow (AF-1: Table Concurrently Claimed / Lock Expired):

- **3a / 4a.** Another diner books the selected table simultaneously, or the 10-minute hold timer expires before checkout:
 1. System alerts the diner that the selected table is no longer available.
 2. System refreshes the 2D floor plan with updated availability.
 3. Diner is prompted to pick another open table while retaining their current food cart.
 4. Use case resumes from Step 3 of MSS.